

sophisticated simulator developed by the team, which takes as input several location-specific parameters such as the required power level, temperature profile, grid availability profile, and relative costs of DG, solar PV and battery power, and uses that to arrive at an optimal power backup plan for that site.

The solution is practical, flexible and future-proof. Operators can use it to not just optimise capital and operational expenditure but also compare the performance of implementations at various locations, retrofit older base stations to reduce power requirements (especially for cooling), perform what-if analysis to see how the optimal mix would change when grid supply changes, and so on.

In the process of writing this report, the team took hundreds of typical base station configurations into account. This exercise not just validated the efficiency of the simulator but also helped the team to come

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up with some broad recommendations about the proper mix of power options, the way ahead and aspects that need to be improved in the coming years.

Apart from suggesting the optimal mix for various common scenarios, the authors of the report also suggest the

future course of action to be taken by the industry. Specifically, they recommend innovation in BTS equipment to reduce total energy consumption, systems for measuring charge of batteries, charge controllers, remote management of sites, DG sets with governors that enable higher efficiency at lower loads, solar temperature-control equipment, new battery technology and robust solar equipment.

Dr Jhunjhunwala says, "The simulator is not country-specific, though the user-cases discussed in the report are. As and when there is a requirement, new features will be added." He also confirms that they would be licensing the simulator tool for use by third parties at a low cost. They are yet to finalise the terms for the same.

The complete report is available online at http://tcoe.in/download/1/Download-Section_190/1/1.html ●

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